CHAPTER 17

FUTURE POSSIBILITIES

NO ONE CAN ACTUALLY FORETELL THE FUTURE in education, science, and technology with precision. There are too many variables involved, and with the advent of newer developments there follows an exponential rise in the learning curve. Therefore we can only extrapolate change based on current trends. Although this method has its limitations, it is the best that we have at present. The future will generate its own values.

The following projections of the future have little to do with the current scenarios popularized in mainstream publications: gadgets and gimmicks available only to upscale, high-income households such as high-tech kitchens with appliances that "talk" and "think"; more advanced and sophisticated weapons, warplanes and ships; and enhanced personal security systems. All these are relevant to a scarcity-oriented culture, and the need, even the desire for many of them disappears with the implementation of a resource-based economy.

One of the major developments of the future will be the field of Informatics, the science of relevant information. We are already moving from data access to management of information. The Internet and information technology allow us to create and use "stateless" information — new information we create by combining data and information from separate data systems and web sites. Development is also ongoing in knowledge management, although most efforts focus on archiving documents and processes. In a monetary-based society, this is logical. These are "seeable" products. True knowledge management allows unconstrained and simplified access to vast amounts of pertinent information.

In addition, nanotechnology shows enormous potential. Nanotechnology combines optics with lasers. This technology will enable us to assemble matter, atom by atom, into whatever molecular configuration is needed. Even today a variety of micro machines, some far smaller than a grain of sand, are part of our technology. This technology is capable of propelling a tiny turbine made of a silicone compound. When a laser illuminates the turbine and the beam is focused at the turbine blades, these micro machines spin rapidly and can be used for many different purposes.

Other micro machines would be used to clear plaque from blood vessels and perform pre-programmed surgical procedures. Eventually medical nano-replicators may replace damaged or non-functioning organs. Some may even outperform the replaced organ. This

includes livers, hearts, eyes, brain tissue, and more. Nanotechnology will lead to a sub-microscopic revolution, not only in the field of medicine but in industry as well.

In the industrial sector production machines would become much more versatile; dies would be programmed to assume any required configuration by varying the molecular bond, while maintaining the accuracy of the system throughout the production process. Each machine becomes faster and more versatile, performing an almost unlimited range of tasks. Noise abatement systems will be used throughout the industrial environment.

Eventually, the need for the transportation of goods and services would also diminish. Products could be replicated and produced within one's own community, eventually within one's own home.

With other forms of energy we would be able to explore outer space. Intelligent robots and mega machines would "terraform" (that is, modify) the uninhabitable planets above ground and underground to support human and plant life and provide all the necessary conditions to sustain human colonization.

The replacement of paperwork by computerized technologies enabled industries to save thousands of feet of the space formerly used to house documents. This also eliminated thousands of clerks and secretaries. Microchip technology freed more than seventy percent of the storage space formerly needed. At present, millions of people throughout the world access electronic information storage systems from their homes or from anywhere else they may be. These information storage systems will continue to shrink, especially with the advances in nanotechnology. What may occupy thousands of square feet of storage space today with molecular information storage systems will be able to fit on the head of pin.

People could have microscopic implants so that, in the event of an accident, when they arrive at a hospital the implant could instantly download all the relevant medical information about the individual. This would eliminate all emergency room paperwork and make the diagnoses much faster and easier to perform.

Another fascinating process is advanced shape memory in plastics, metals and other materials. This process will probably lead to sub-microscopic, electro-morphic materials. Such materials will be able to alter their external appearance to whatever shape will produce the best performance. Most home furnishings in the future would be capable of adjusting their shapes to accommodate the human body.

Other aspects of the future would be realistic, 3-D visualization with teletactile and olfactory enhancements that could enable one to touch and smell flowers or any other visual representations, from the undersea world to the stars.

With the advent of artificial intelligence, the technological performance of machines will outpace/outperform the need for managerial oversight. Molecular circuitry will eventually provide the necessary interface to enable human beings to engage in an intelligent discourse with a machine. This technology would enable the machine to repeat and understand any written or spoken language in the world — including sign language or Braille — and

permit instantaneous translation throughout the world. These same implants could be capable of doing research in all branches of the physical sciences.

Not only will this newer technology replace humans in the production sequences, but in the service sectors as well.

The introduction of this and other computer-generated technologies for multi-media application could affect the future of entertainment, resulting in 3-D, teletactile and olfactory images that simulate living beings and locations. The results could be so life-like, it would be almost impossible to separate simulation from reality.

With teletactile simulation in the projection of a human image, we will actually be able to shake hands with our virtual visitors and even walk with them through our gardens. These virtual visitors would be able to "pick up" objects and examine them. They would appear not as synthetic projections but as living, breathing, human beings.

Today we can only imagine what teletactile imaging could mean to people who have lost loved ones or who have lost limbs or their eyesight. It also opens the possibility of being in many different places at the same time, a scenario once considered an absolute impossibility. This occurs on a very primitive level today through the electronic media, when the president addresses the nations of the world.

If this boggles the minds of forward-thinking people, think what it could do to our culture-bound, rigid notions of reality. Regardless of our personal views regarding the worth or the value of this technology, it is coming. We already live in a world where yesterday's fantasies have been surpassed by today's realities.

Probably the only thing we can know for sure about the future is that it will be very different from the world today. But whatever difficulties we have trying to understand life in the future, it is nothing compared to the difficulty people of the future will have understanding the way we do things today. They will most likely find it hard to believe that human beings could have organized themselves in such an absurd fashion into nations, after which they set about using scientifically-designed weapons to slaughter each other. As they watch movies of the past they will probably be astounded at the tobacco smoke emanating from people's nostrils and the ostentatious clothing and omnipresent jewelry. They will find our simple, animal-like emotions of hostility, rage, and jealousy incredible. People in the future will probably not look back with nostalgia to a world threatened by atomic oblivion, environmental degradation, and economic and political activities permeated by greed and self-centeredness. How simple, how crude and pathetic we will appear in the eyes of our descendants — perhaps just as strange and unpleasant a sight as that of our own imaginings about *our* primitive ancestors.

When biological technology becomes further advanced, human beings as we know them will become a modified species. If we as human beings fail to include the possibility of this development in our overall, social evolution we will witness the decline of our species. All aspects of social innovation must be able to allow for change in a constantly evolving world.

CHAPTER 18

THE OCEAN FRONTIERS OF TOMORROW

THE WEB OF LIFE ON OUR PLANET is supported by the hydro-cycle, that great variation of forms of water which are part of the planetary circulation: the oceans, snow, ice, rain, lakes, groundwater, and aquifers. This constantly renewed circulation, powered by the heat of the sun, the rotation of the earth, and Coriolis forces, supports the entire life cycle, including humankind.

We often speak of underdeveloped land areas, but rarely of the greatest undeveloped natural resource on the planet — the world's oceans. Although humans have used the oceans of the world for thousands of years as a source of food and transportation, we are only now beginning to recognize the enormous potential and diversity of this relatively untapped resource. The oceans offer an almost limitless environment for food, energy production, minerals, pharmaceuticals, and much more.

The ocean is the only resource that has kept ahead of the population explosion. And yet we see little regard for ocean life, which is essential to all life on Earth. In August 1970, the U.S. Army deliberately dumped containers holding 67 tons of nerve gas into the Atlantic Ocean. Add to this the fact that the dumping ground was close to a main artery in this life support system, the Gulf Stream. The navies of the world, the fishing fleets, cruise lines and many coastal cities casually use the ocean as both trashcan and toilet.

If intelligently-managed, the oceans could easily supply more than enough resources to feed the world's hungry. Billions depend on the sea for their primary source of protein. Life is abundant and varied here. But mismanagement of run-off created huge, lifeless areas in the Gulf of Mexico where the Mississippi River drains. Major marine species and the coral reefs that nurture them are rapidly disappearing — not naturally or because their death in any way prolongs our way of life. On the contrary, these extinctions endanger us and derive from our own arrogance and ignorance. Even amid the most complex, living ecology, we act as predators rather than the symbiots we are.

Although the overwhelming majority of sea life dwells near the surface, in the chill, murky depths miles below, where even sunlight never ventures, life abounds despite fantastic pressures and temperatures. In near-freezing temperatures, boiling vents of toxic gasses support a wide array of sea life we have only begun to study.

Great rivers cross the oceans of our planet, set in motion by the earth's rotation. These immense, oceanic currents travel at varying speeds, at different depths, and even in opposite directions. It has been estimated that the Gulf Stream carries about 30 million cubic meters of water per second past Miami, Florida. This is more than five times the combined flow of all the fresh water rivers of the world. If this potential energy were harnessed, it is estimated that the project would deliver close to a thousand million watts on a 24-hour basis or as much as 2 large nuclear plants, without environmental contamination or radiation danger.

In addition powerful winds, waves, and currents provide enormous potential for the generation of electric power. The vast potential of the seas can also be realized by the production of energy crops from biomass by converting waste organic materials into gaseous or liquid fuels.

On the sea floors, and in the brine-filled waters themselves, we find a vast storehouse of metals and minerals that may be used to help resolve resource shortages. However, "harvesting" the metals and minerals will require new technologies that do not disturb the fragile sea floor.

CITIES IN THE SEA

Colonization of the oceans is one of the last frontiers that remain on earth. In the redesign of our cities, prodigious oceanic city-communities may evolve. To fully utilize this bountiful wellspring of resource, the way of the future embraces the development of large marine structures designed to explore the relatively untapped riches of the world's oceans. These cities in the sea could provide improved mariculture, fresh water production, power, and a variety of mining, which could offset land-based mining shortages. They could provide almost unlimited riches in pharmaceuticals, chemicals, fertilizers, minerals, oil, natural gas, sweet water, and tidal and wind power, to name only a few. Ocean-based and space-borne sensors will constantly track tidal flow, marine life, water composition and temperature, atmospheric conditions, and a myriad other vital signs.

Ultimately the development of these ocean communities will greatly relieve the land-based population pressures. The population of such cities would vary from several hundred to many thousand and would be located throughout the world. These will all be controlled, managed and operated by automated systems and be part of the international communications networking.

USE

Some of these cities can serve as universities and research centers where students from all nations of the world study marine sciences and management. They could also serve as monitoring stations of ocean currents, weather patterns, marine ecology, pollution, and geolog-

ic phenomena. For additional marine exploration, robotic submersibles would be available. The oceans are, after all, essential to our survival and a critical part of the earth's carrying capacity.

Other sea platforms can be used as rocket-launching systems. Space vehicles launched at the equator save a great deal of energy because the equator is the fastest moving portion of the earth. Locating launch sites there takes full advantage of the earth's rotation for additional thrust. This requires fewer thruster burns to reach geocentric orbit, or the orbit whereby a satellite rotates with the earth and remains in a stationary position relative to it.

For polar orbits, the launch platforms could be located off the West Coast of the U.S. The computerized control and command systems could be located on ships or on the platforms themselves.

Still other areas of the oceans would remain largely untouched. These are treasures in themselves and need never be turned over to technological development or exploitation. Their reclamation, enhancement, and preservation should be a priority for global conservation.

The Caribbean and the emerald shoals of the vast banks of Eleuthera feature some of the clearest waters in the Bahamas and one of the most beautiful coral atolls in the Western Hemisphere. The waters surrounding these islands vary in hue from the magnificent deep blue of the Gulf Stream to shimmering shades of green. Similar areas exist in the South Pacific and many other locations throughout the world, where thousands of miles of shoreline remain unmarred by signs of human habitation. In the new spirit of world cooperation, some of these areas could be set aside as international marine parks for the education and enjoyment of all, inviolate from exploitation. In these areas, the only human intervention taken would be measures to preserve and protect these aquatic sanctuaries.

LIFESTYLES ON THE SEA CITIES

Cities of the sea could offer new and fascinating lifestyles for millions of inhabitants, while easing the land-based population pressure. Massive ocean structures would be both above and beneath the sea. These structures would represent a spectacular engineering achievement with aircraft, sea craft and submersible access.

Some of these cities in the sea could also serve as underwater international parks where visitors could observe the great protected reefs of the world. Through huge undersea windows they would be able to view the wonders of this environment in leisure and comfort; from a computerized chair, they could communicate with dolphins and other forms of marine life, while the more daring among them could leave the premises by way of airlocks and go on diving expeditions.

CONSTRUCTION

Cities in the sea could be among the greatest achievements of the twenty-first century. One of the most efficient designs would be a circular configuration, multi-storied and fabricated of steel, with glass of superior strength and pre-stressed concrete reinforced with carbon fibers.

Some may be floating versions while others could be built on pilings, with flotation barriers that would prevent wind and heavy seas from damaging the structures. In deeper waters, the floating platforms could be anchored to the seabed. Other ocean platforms could float freely. These platforms would be self-propelled and extremely stable, ballasted by columns about 20' feet in diameter that penetrate 150' below the surface. In the lower portions of these floating, cylindrical columns are a series of disks that extend out about 6 feet, spaced approximately 10' apart. These disks tend to keep the platform steady in any type of adverse weather conditions. A belt surrounding the entire project or other devices act as a breakwater.

Some of these cities can be constructed in technically developed countries and towed to their destination in sections or as complete operating systems, similar to the manner in which oil platforms are transported to their destinations today. Other configurations can be designed as variable composite structures, assembled on site and modified to serve many different functions. They can also be disassembled and relocated if required. They would vary in size depending on their function and could be as large as three-quarters of a mile in diameter with flexible interconnections.

Other above-surface structures anchored to the seabed could serve as efficient bases for mining operations. These dome-shaped structures would be totally automated, their flotation levels adjusted by flooding or emptying their buoyancy chambers. They may be constructed on land and towed to their destination, where they can be submerged and anchored in place. From the tops of these structures, a cylindrical concrete conduit will extend 150 feet above the ocean surface, its surface-base encircled by a floating dock system which would rise and fall with the tides and accommodate both surface and submersible craft.

All marine development must be in full accord with the total carrying capacity and sustainment of the ocean environment. *Before initiating any of these major projects, it is imperative to take into account the possible negative impact on the entire hydrosphere — the rivers, estuaries, lakes and oceans*

ENERGY

On these and other floating cities or platforms powerful wind turbines could capture the ocean breezes. Solar and wind power generators would be located on most of the upper decks. Cold water from the depths of the ocean can also be pumped up for various uses such as the conversion of this heat differential into electrical energy. This process could provide a continuous supply of electricity far in excess of the cities' needs.

With our dwindling agricultural space, ocean farming offers an area for unlimited energy crops. Biomass and processing of waste organic materials into liquid or gaseous fuel would provide additional energy from the fermentation processes.

MARICULTURE

Mariculture, the planned cultivation of marine crops, and fish farming communities can be designed to support more than one type of marine life. Many of these communities could maintain a balance of species in a mutually supporting, symbiotic relationship, while emulating natural conditions as closely as possible.

A wide variety of aquatic plants could also be cultivated in multiple layers, and suspended by cables in underwater fields adjacent to these cities. In some instances the tops of the plants could be harvested automatically, leaving the roots and the lower third of the plant to grow new crops without replanting.

These floating ocean platforms could also be equipped with solar-operated desalinization plants, which extract fresh water for hydroponics farming and other uses. Upwelling can also be harnessed to extract deep-sea nutrients to supply aquaculture farming. Of course, any attempt at aquaculture or mariculture must be subject to the international monitoring of ocean farms.

This would not only provide fish farming complexes, but could introduce the most advanced principles of poly-culture to maintain the reproduction and natural balance of species. Every precaution must be taken to avoid disrupting or spoiling the very spawning grounds that have sustained the human race for countless centuries.

TRANSPORTATION

Immense, floating structures would be equipped with loading and docking facilities for vessels. Huge ships that serve as processing plants could transport passengers and freight to these cities in the sea.

The upper deck would be equipped with a landing area for helicopters or VTOL aircraft. Computerized lift-units could facilitate vertical, horizontal and radial travel within these structures.

JOINT VENTURE

Ocean cities will help to provide additional food and resources for the needs of the world. Cities in space, on land, and in the oceans can be managed by a global resource management system, thus serving every nation of the world while maximizing the well-being of everyone.

Where a project of this magnitude is concerned, it is imperative that its benefits be shared equally by the entire global community. Eventually, mineral wealth of the oceans and the other resources of our world must be shared by all nations as the common heritage of humankind. If we fail to adopt these measures, the rapid exploitation and deterioration of the ocean's resources may not be reversible.

OFF SHORE APARTMENTS

Offshore apartment buildings of concrete, steel, glass, titanium, and a wide variety of new synthetic materials can be built to relieve the population pressure in areas like Hong Kong, Tokyo, Los Angeles, and New York. The materials used in such projects would be engineered to withstand the corrosive effects of the harsh ocean environment.



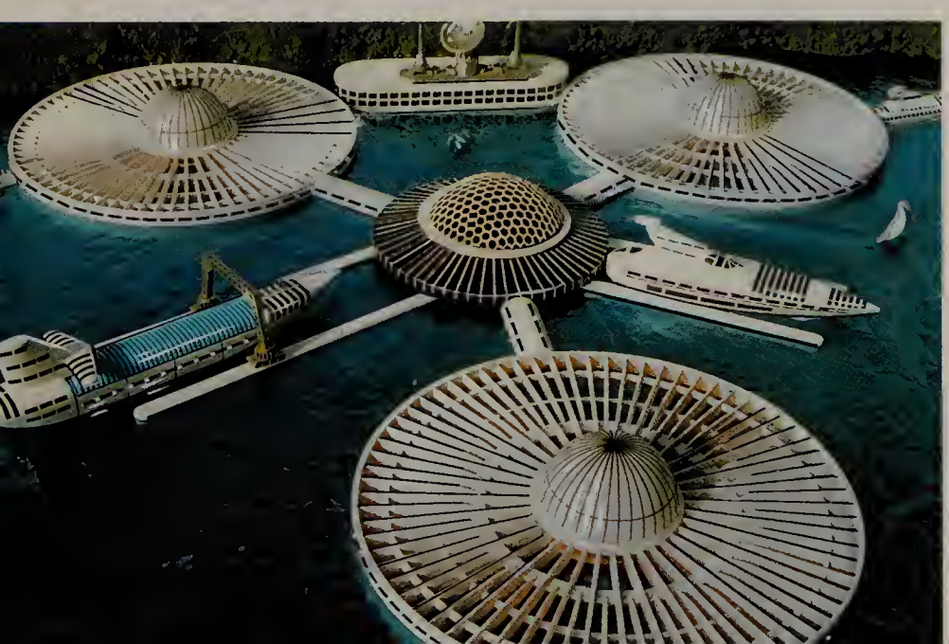






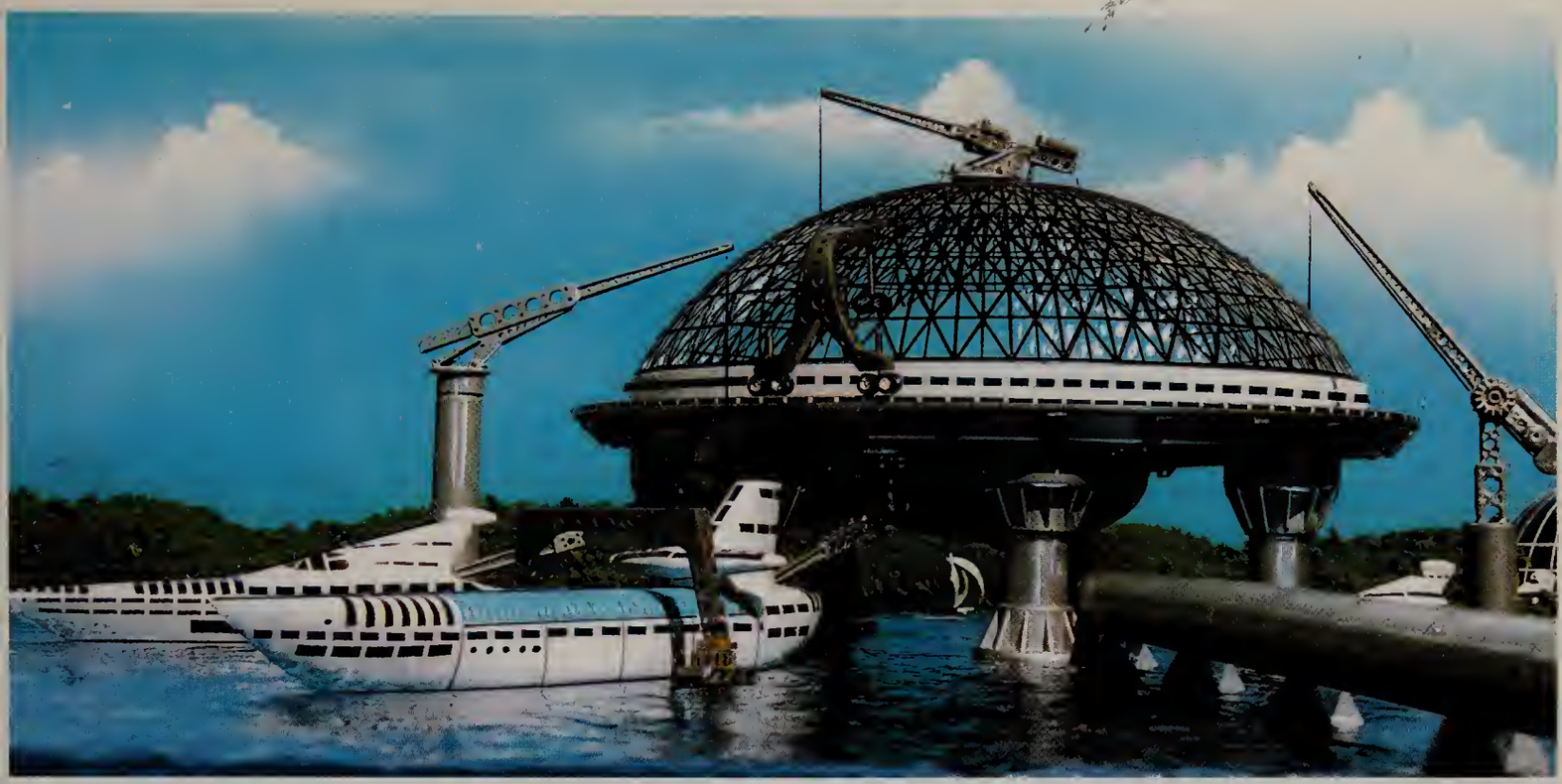
ARTIFICIAL ISLANDS IN THE SEA

This artificial island in the sea is designed to serve the oceanographic sciences. Multiple docking and landing facilities for VTOL aircraft surround the entire island. Water-based recreation will be a part of life in these water-borne communities. People will be able to participate in research, sailing, scuba diving, and many other surface and sub-surface activities without disturbing the balance of the marine environment.



MARICULTURE & SEA FARMING

This is a conceptual view of a mariculture and sea-farming system. Such systems cultivate and raise fish and other forms of marine life to help meet the nutritional needs of the world's people. Capable of cultivating a great variety of marine life, these structures would be equipped to permit the free flow of water throughout. They are designed to be a non-contaminating, integral part of the marine environment.

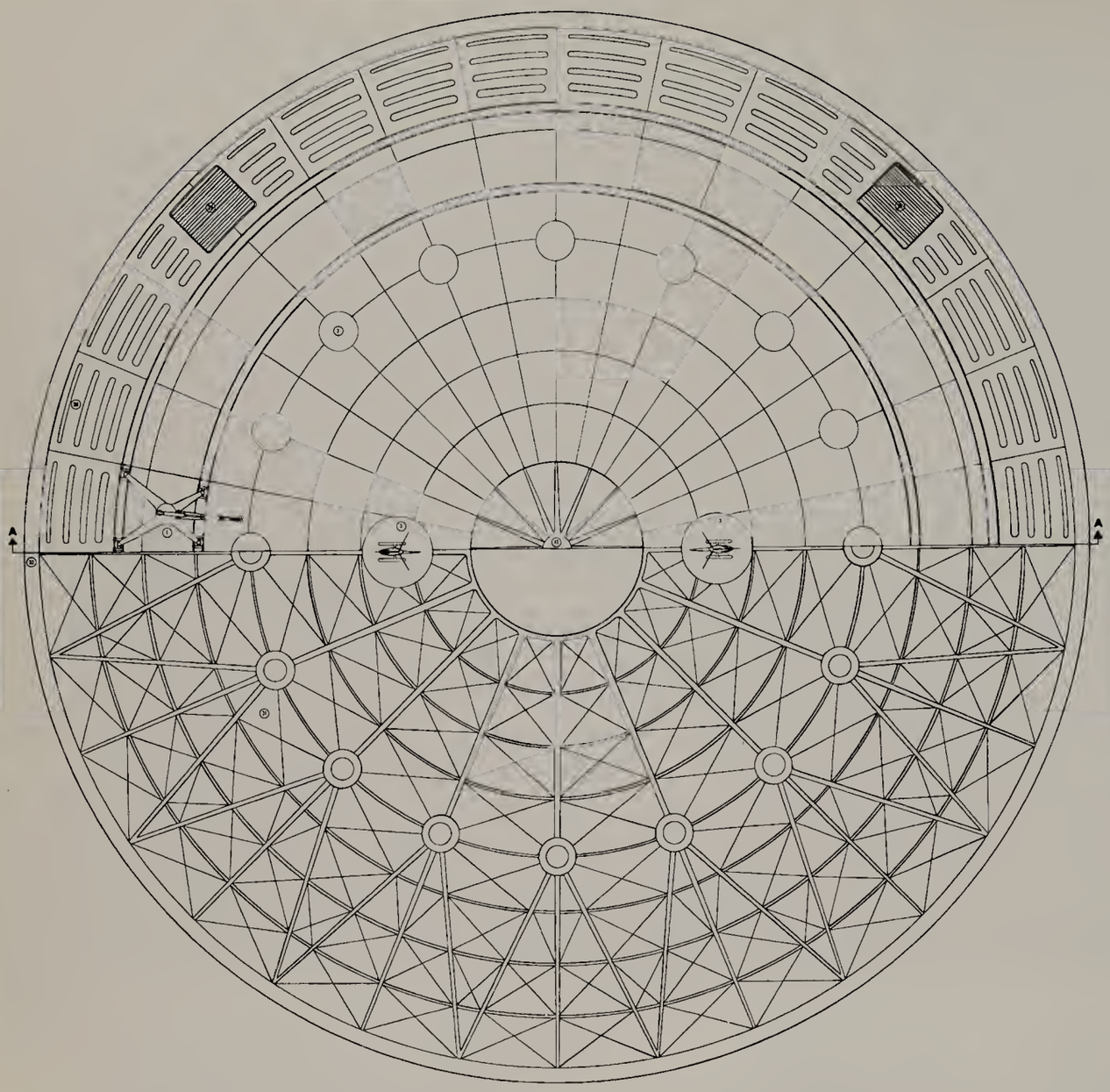


OCEAN MINING MEGA-STRUCTURES

These "cities in the sea" provide improved mariculture, fresh water production, power, and deep-ocean mining, which can help alleviate shortages of land-based minerals. Such structures could provide us with almost unlimited resources: pharmaceuticals, chemicals, fertilizers, minerals, metals, oil, natural gas, drinking water, ocean farming, as well as tidal and wind power and more.

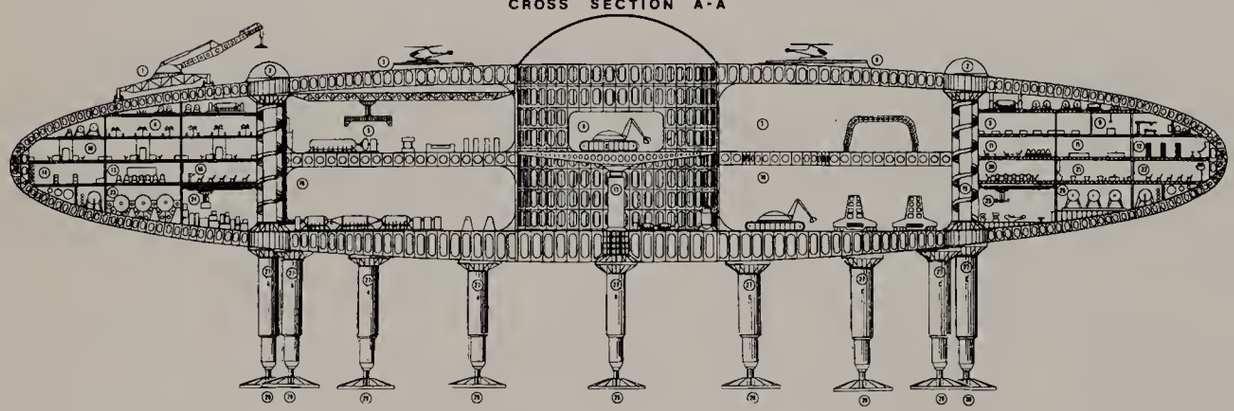


SOCIOCYBERNEERING



TOP VIEW

CROSS SECTION A-A



① CRANE
② ELEVATOR AND STAIRWAY
③ HELICOPTER PAD
④ ENGINEERING AND DESIGN DEPARTMENT
⑤ ASSEMBLY MACHINE SHOP AND SERVICE AREA
⑥ FREIGHT ELEVATOR
⑦ PRODUCTION JIGS ASSEMBLY DECK
⑧ HOSPITAL
⑨ OPERATING ROOM AND X-RAY DEPARTMENT
⑩ STAFF QUARTERS
⑪ LOUNGE AND RECREATION AREA
⑫ LIBRARY
⑬ CONFERENCE ROOM
⑭ COMPUTER ROOM
⑮ PROJECT ORIENTATION ROOM
⑯ POWER GENERATORS AND CONTROL ROOM
⑰ HYORAULIC LIFT ELEVATOR
⑱ MACHINE ENGINEERING DEPARTMENT
⑲ STAIRWAY AND ELEVATOR SHAFT
⑳ CONFERENCE ROOM
㉑ LABORATORY
㉒ FILM ROOM
㉓ PUMPING STATION AND FUEL STORAGE TANK
㉔ MATERIALS STORAGE
㉕ MACHINE SHOP
㉖ FLUID STORAGE SYSTEMS
㉗ HYDRAULICALLY OPERATED UNDERWATER SUPPORTS
㉘ SELF LEVELING PLATE
㉙ FREIGHT SERVICE OPENINGS
㉚ SOLAR GENERATOR SYSTEMS
㉛ STRUCTURAL FRAMING
㉜ PERIMETER STRUCTURAL RING

PRELIMINARY DESIGN STUDY FOR FLOATING MEGASTRUCTURES

DESIGN: J. FRESCO DRAWN: R. MEADOWS DATE: 12 - 9 - 77 SCALE: 1/2"=1'-0"

CITIES IN THE SEA

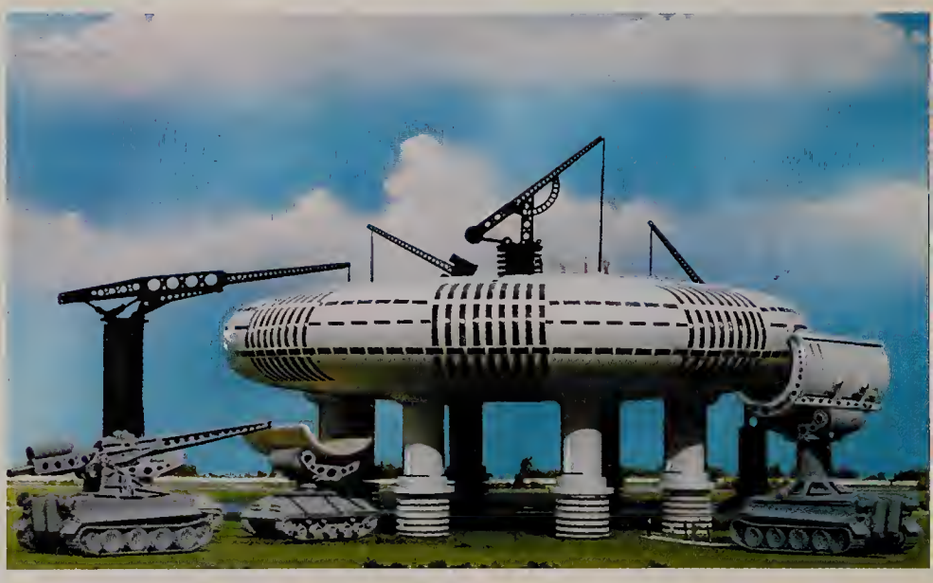
Thousands of self-sufficient cities in the sea, varying in design according to their location and function, will eventually alleviate land-based population pressures. Some will serve as oceanographic universities to survey and maintain a dynamic balance in the oceanographic environment.





CONSTRUCTION OF A FLOATING MEGA-STRUCTURE

This illustration depicts a "floating mega-structure" that is being assembled in a dry dock entirely by robotized, automated systems free of any human intervention. After construction, these structures could be towed in sections or as completed units to locations where they would be anchored to the seabed. In some instances these cities could even travel and relocate.



CITY IN THE SEA WITH A MODULAR FREIGHTER

This modular freighter, leaving a city in the sea, consists of detachable sections that can be rapidly loaded or unloaded. The number of sections varies, depending on the amount of freight to be delivered. When all the modules are connected they can be propelled as a single unit. When the freight arrives at its destination, the selected modules can be disconnected and towed to docks.

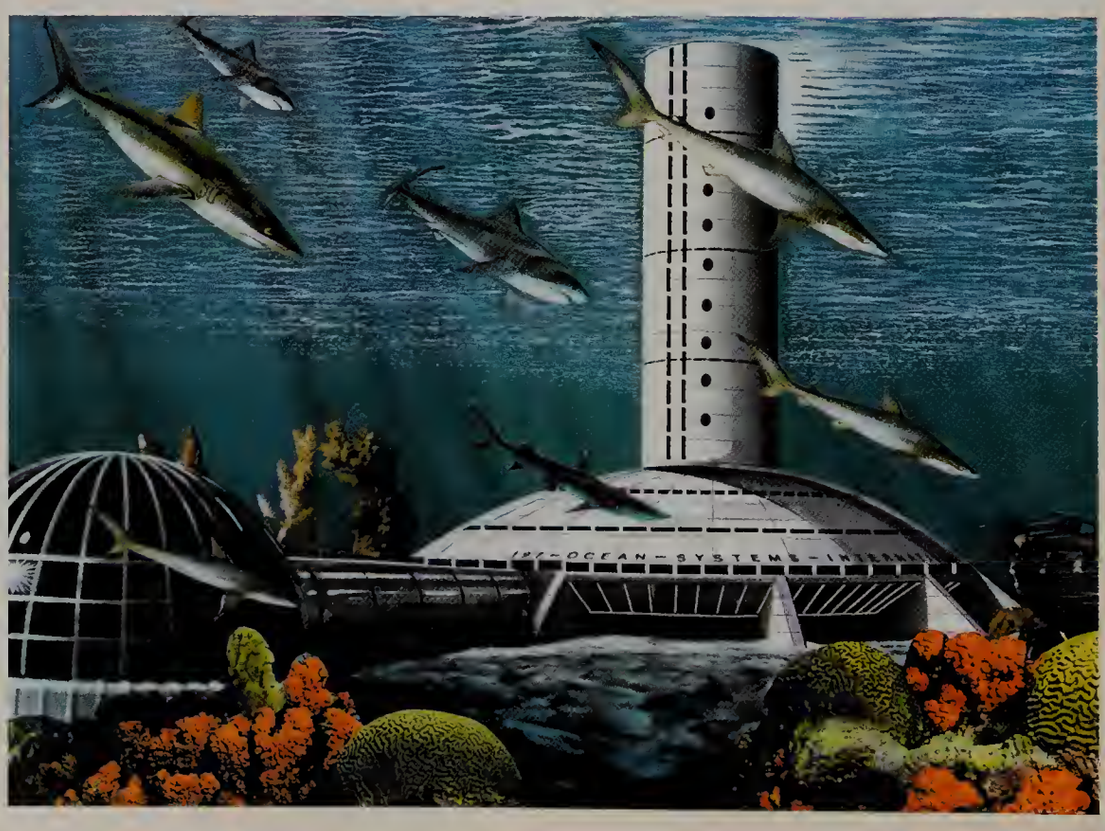


FLOATING SEA DOMES

These unsinkable, floating sea domes will house those who prefer unique offshore or island living. In the event of severely inclement weather, they could easily be towed ashore, mounted and locked to elevated support structures. They are all equipped with retractable enclosures that cover the outer deck.

UNDERSEA OBSERVATORIES

Elevators transport visitors to underwater oceanographic viewing or research facilities. An extended under-water tunnel connects to other aquatic facilities and provides expansive, panoramic viewing of the undersea world in its natural habitat without disturbing the ocean environment.



CHAPTER 19

BEYOND UTOPIA

IN 1898, EDWARD BELLAMY wrote the book *Looking Backwards.* It presented a social system with many advanced ideas for its time. This best seller generated a great deal of interest and many people wrote to ask how the type of cooperative society Bellamy envisioned could be brought about. But our nation at that time was not prepared for a transition of this magnitude.

The proposals he presented, and those of Plato's *Republic,* the writings of Karl Marx, H. G. Wells in his book *The Shape of Things to Come,* and many others all represent attempts to find workable solutions to the many problems that earlier civilizations have left unresolved. There is little doubt that at the time of Bellamy's books, social conditions were abominable, which made the Utopian ideal much more appealing. What appears to be lacking in most of these concepts, however, is any type of plan or process to facilitate the transition. Most of the early visions of Utopia did not allow for changes in either technology or human values, tending to arrest any innovative efforts. And all have lacked a set of blueprints and methodology for implementing the ideas in a comprehensive form, and a competent staff to effect the transition.

Now, at last, we have such a vision and the means to make it a reality. In recent times we have evolved the necessary technology to surpass the fondest hopes and dreams of any social innovators of the past. Although many of the concepts presented in this book may appear to people of the early twenty-first century to be unattainable goals, all of these concepts are based upon known scientific principles. The only limitations upon the future of humankind are those that we impose upon ourselves.

The social direction we speak of has no parallel in history with any other previous political ideology or economic strategy. Just because previous attempts failed, however, is not an acceptable reason to stop trying. The real danger lies in doing nothing.

Determining the parameters of this new civilization will require a break with many of the traditions of the past. The future will evolve its own new paradigms that will hopefully fit each, successive phase of human development.

Perhaps a major influence on the origins of the earliest Utopian concepts were some of the world's religious teachings. In these imagined visions of heaven there were no property lines, banks, money, police, prisons, militarism, or private ownership.

Not so many years ago an attempt was made in the U.S. to understand a social system very different from our own. A film called *"The March of Time"* had this to say about Soviet Communism: "We believe that the American free-enterprise system will function better than the collective system. However, we wish you the best of luck on your new and unusual social experiment." The failure of communism to provide for human needs and to enrich the lives of its citizens is not unlike our own failures. In all established social systems it is necessary to devise different approaches to improve the workings of the system.

Thousands of failures occurred before the first workable airplane was produced. Dr. Erlich attempted over 606 different approaches to controlling syphilis before one finally proved successful. Some of the technology we use today, such as televisions, radios, aircraft, and automobiles are in a constant state of improvement and modification. Yet our social system remains largely static.

An inscription on one of our government buildings reads as follows: "Where there is no vision the people perish." The major reason for resisting change is that it tends to threaten the vested interests. The fear of social change is unfounded, because the history of civilization is one continuous experiment. Even the American free-enterprise system, during its earliest stages, faced a multitude of problems even more severe than they are today — long hours, exploitation of child labor, inadequate ventilation in industrial plants, lack of rights for women and black Americans, hazardous conditions in mines, and racial prejudice. Despite its many problems, it was the greatest attempt in history in terms of innovation in lifestyle, architecture, technology, and the general pursuit of progress. All we recommend is that we continue our process of social experimentation to transcend the limitations of our current society and enhance the lives of everyone.

Our future does not depend on present-day beliefs or social customs, but will continue to evolve a set of values unique to its own time. There are no Utopias. The very notion of "Utopia" is static. However, the survival of any social system ultimately depends upon its ability to allow for change to improve society as a whole.

CHAPTER 20

THE VENUS PROJECT DIRECTION

THE VENUS PROJECT IS AN ORGANIZATION based on the ideas, designs and direction presented in this book, representing the life's work of its originator and Project Director, Jacque Fresco. Its 25-acre research and design center is located in Venus, Florida, where the future is taking shape now. *The Venus Project's* purpose is to design, develop, and prepare plans to build the first experimental city. Here we are developing alternative energy systems, city designs, transportation, manufacturing methods, and more. Along with this research we are creating blueprints, renderings and models, while holding seminars and producing books, videos and other written material to introduce people to the aims of *The Venus Project* and our vision for the future.

Without reservation, we conclude that *The Venus Project* cannot be accomplished within a monetary-based society. It would take too many years for any significant changes. Any attempt to adapt our goals to a monetary-based system would water them down to such an extent that the changes would be insignificant.

As emphasized earlier in the book, it is no longer the repetitious work of just the laborers that automation will continue to phase out, but also that of engineers, technicians, scientists, doctors, architects, artists and actors as well. All of these professions will be a short-lived means of providing purchasing power. It is simply a matter of time before automated systems provide nearly all necessary services and products. Virtually all tasks presently performed by human intelligence could be performed by automated systems. This need not be a cause for alarm. In a cybernated, resource-based culture, human work will be involved in creative endeavor and problem solving. There appears to be no limit to the services and benefits for all human beings that can be performed by computerized technologies and *AI.*

The Venus Project calls for a straightforward approach to the redesign of a culture, in which the age-old inadequacies of war, poverty, hunger, debt, environmental degradation and unnecessary human suffering are viewed not only as avoidable, but totally unacceptable.

If we are sincere and genuinely concerned with resolving most of our problems, we must strive towards having the Earth's resources declared as the common heritage of all the world's people. For the reasons discussed in this book, anything less will simply result in a continuation of the problems inherent in our present system.

To effect a transition from our present, politically incompetent, scarcity-oriented and near-obsolete culture to a more humane society will require a quantum leap in both thought and action. Until recently, change came slowly. One group of incompetent leaders merely replaces another. The problems we face today cannot be solved politically or financially. Our problems are highly technical in nature and require fundamental changes in our thinking and values. There is not even enough money available to pay for the required changes, but there are **more than enough resources**. This is why *The Venus Project* advocates the transition from a monetary-based society to the eventual realization of a resource-based world economy.

The money-based system evolved centuries ago. All the worlds economic systems — socialism, communism, fascism, and even our free enterprise system — perpetuate social stratification, elitism, nationalism, oppression and racism, based primarily on economic disparity. Power relates to an individual's or group's ability to withhold food, shelter, health care, education and resources from the poor and disadvantaged. The basic sustaining factors of life are held hostage for hours of labor, as represented by a salary. Consequently, as long as a social system uses money or barter people and nations will seek to maintain the economic competitive edge; if they cannot do so by means of commerce they will attempt to do so by means of boycotts, blockades, or military intervention. Even at the time of this writing, three major economic boycotts and multiple military interventions are in progress around the world.

All social systems, regardless of political philosophy, religious beliefs, or social customs, ultimately depend upon natural resources, clean air and water, arable land, and the necessary technology and personnel to maintain and secure a high standard of living.

The Venus Project concludes that the Earth is abundant with plentiful resources. Our practice of rationing resources through monetary control is irrelevant and counter-productive to our survival.

Modern society has access to highly-advanced technologies that can provide sufficient food, clothing, housing, medical care, education, and the development of a limitless supply of renewable, non-contaminating energy sources. We have the technology, resources and personnel for everyone to enjoy a very high standard of living with all of the amenities a prosperous, global civilization can provide. *This can be accomplished through the humane and intelligent application of science and technology, based upon the existing carrying capacity of the Earth.*

The Venus Project posits that the necessary technology already exists to begin making maximum resources available and to provide food, clean air and water, comfortable housing and transportation, quality health care, environmental stability, and unlimited opportunities for personal growth to all people, not just a selected few.

Our understanding of the applications of technology suggests the possibility of eliminating scarcity by applying renewable sources of energy; humane and intelligent resource management along with the infusion of cybernation will help maintain a balanced-load economy and a far more equitable distribution of the world's resources.

Machines will monitor production and delivery of goods and services, and maintain the protection of the global environment. They will not monitor people. In a resource-based economy, monitoring people is not only socially offensive but also counterproductive.

In a saner and more humane civilization, competent use of machines would shorten the workday, increase the availability of goods and services, and lengthen vacation time. If we utilize new technology to raise the standard of living for all people, the infusion of machine technology would no longer be perceived as a threat. We would work towards replacing most forms of human labor, in its place providing challenging, interesting pursuits in problem solving and creative effort, while encouraging individuality and new incentives. The purpose of all this organized technology is to free people from the monotonous and boring work-a-day world and to enable them to pursue a more meaningful life.

Today, financial barriers place enormous limitations on innovation, development, individual creativity, and incentive. In the world envisioned by *The Venus Project*, people are free of the constraints of profit and control, and are able to explore new dimensions in human existence and pursue knowledge in the arts, sciences, or any other areas of their choice. There is a tremendous, undeveloped potential in all human beings, which is not nurtured in a monetary-based society. All would benefit by the fruition of new ideas. In a resource-based society, the measure of success would be the fulfillment of one's individual preferences and pursuits rather than the acquisition of wealth, property, and power.

HOW WE GET FROM HERE TO THERE

The Venus Project is in the process of introducing a set of values and procedures that may enable us to achieve social nucleation. Our project will provide the designs and blueprints for a prototype community to test the validity of our proposals; we will strive to achieve a relevant orientation by which people may adapt intellectually and emotionally to our new technological age. We feel that anything short of overall social design would be inappropriate and ineffective.

Since we begin in a monetary-based society, the means for raising funds to construct this experimental community can be accomplished in several different ways.

One is through the production of a major motion picture depicting the advantages of this new social system for all the world's people. Another possibility is the building of a theme park where visitors would actually experience some of the many benefits of *The Venus Project's* proposals. Books, videos, blueprints, models, a movie script, and the 25-acre research and development center have already been completed.

Any funds raised by these proposed projects, along with contributions, publications, videos, seminars, lectures, and grants, will be used to help initiate and construct the first experimental city.

Our proposals are submitted to the general public and to all educational institutions. We invite your participation. If enough people find the proposals acceptable and join with

us in this new advocacy, we could form the nucleus of an organization to more forcefully implement the aims of *The Venus Project.*

EXPERIMENTAL CITY

The Venus Project proposes the building of a new, experimental city, the purpose of which is:

1. To be a laboratory to test the validity of the project's designs and proposals.
2. To establish a permanent center that could be used for future long and short-term planning.

The circular configurations of these new cities proposed by *The Venus Project* are not just stylized, architectural conceptualizations, but the results of years of research into an environment that best serves the needs of the occupants efficiently and economically. Without sufficient knowledge of the symbiotic interrelationship between humanity and the environment, it would be extremely difficult to develop workable solutions to our many problems. *The Venus Project* has taken this and many other factors into careful consideration and study.

This new experimental city would be devoted to working towards the aims and goals of *The Venus Project,* which are:

1. Realizing the declaration of the world's resources as being the common heritage of all the earth's people.
2. Transcending the need for the artificial boundaries that currently and arbitrarily separate people.
3. Replacing the monetary-based economy with a resource-based world economy.
4. Assisting in stabilizing the world's population through education and voluntary birth control.
5. Reclaiming and restoring the natural environment to the best of our ability.
6. Redesigning our cities, transportation systems, and agricultural and industrial plants so that they are energy efficient, clean, and able to conveniently serve the needs of all people.
7. Gradually outgrowing the need for corporate entities and governments (whether local, national, or supra-national) as a means of social management.
8. Sharing and applying new technologies for the benefit of all nations.
9. Developing and using clean, renewable energy sources.
10. Manufacturing the highest quality products for the benefit of the world's people.

11. Requiring environmental impact studies prior to construction of any mega projects.
12. Encouraging the widest range of creativity and incentive toward constructive endeavor.
13. Outgrowing nationalism, bigotry and prejudice through education.
14. Eliminating any type of elitism, technical or otherwise.
15. Arriving at methodologies by careful research rather than random opinions.
16. Enhancing communication in schools so that our language is relevant to the physical conditions of the world.
17. Providing not only the necessities of life, but also offering challenges that stimulate the mind while emphasizing individuality rather than uniformity.
18. Finally, preparing people intellectually and emotionally for the changes and challenges that lie ahead.

Like all innovative social proposals, ours starts with a few devoted people who dedicate their time to informing others of the humane benefits of this new direction and to a variety of tasks pertinent to the project. All people are invited to participate in whatever capacity they can, to help carry out the initial design phases of this new experimental city. During the initial phase, we will utilize an interdisciplinary team of systems analysts, engineers, computer programmers, architects, city planners, sociologists, psychologists, educators, etc.

The design of *The Venus Project* does not regard environmental conditions as fixed or static. We must allow for adaptation and change as a continuous process. This avoids the tendency to perpetuate temporary arrangements beyond their period of usefulness.

A circular city would be a transitional phase and could evolve from a semi-cooperative, money-oriented society to a resource-based economy. This could be the prototype for a series of cities to be constructed in various places throughout the world. The rate of progression will depend upon the availability of funds raised during the early stages and the people who identify with, participate in, and support the aims and direction of *The Venus Project.*

As these new communities develop and become more widely accepted, they may very well form the basis of a new civilization, preferably through the process of evolution rather than revolution.

We are well aware no one can actually foretell the shape of the future. We can only extrapolate on present information and trends. Population growth, technological change, worldwide environmental conditions, and available resources are the primary criteria for future projections.

We are also aware that there is no single philosophy or point of view — religious, political, scientific, or ideological — that someone would not be able to take issue with. We

feel certain, however, that the only aspects of *The Venus Project* that may appear threatening are those which others project onto it.

The Venus Project is neither Utopian nor Orwellian, nor does it reflect the dreams of impractical idealists. Instead, it presents attainable goals requiring only the intelligent application of what we already know. The only limitations are those which we impose upon ourselves.

The Venus Project does not advocate any type of sabotage of the existing free-enterprise system. We believe that it will come to an end of its own accord, as indicated in this book. We do, however, wish to provide an alternative approach for your consideration. We encourage you to become better informed about the proposals of this project through our books, videos and seminars. If you identify with this direction, we welcome you to join with us and work towards its realization.

A view from space presents us with an ever-changing image of this beautiful planet, an impression that our world is one. The artificial, national borderlines do not appear. We are finally beginning to realize that humankind is but one single family. Only when the nations of the world can agree upon a common direction and can state the problems precisely, will we be able to reach workable solutions. In times to come, the need for all forms of loyalties and beliefs that divide, mislead, and destroy social continuity should vanish, and we may realize that most of the major problems confronting the nations of the world are human in origin. We must act, and act swiftly, to avoid further degradation of both our planet and ourselves. We must act while both the earth and humankind still retain their intrinsic worth.

TO CLOSE, WE INVITE YOUR COMMENTS AND PARTICIPATION.

In the spirit of Angeles Philolias:

"In the final analysis:
We are one people — we share one Planet.
We wish to live in Peace and heal the Earth.
We will do our best to make this a reality.
We will change what needs changing based upon loving, sharing, and serving all humanity.
We, the people of Earth, are one family."

FOR FURTHER INFORMATION PLEASE CONTACT:

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FOR FURTHER INFORMATION

VIDEOS

WELCOME TO THE FUTURE

It is an overview explanation of the direction and the aims of The Venus Project. This video presents an attainable vision of what our world could be if we intelligently apply science and technology with environmental and human concern — a future where war, poverty and hunger could be but a distant memory. It advocates surpassing the monetary system by introducing a resource-based economy, in which all of the world's resources are utilized for the common heritage of all people. It presents a vision of future cities (on land and sea), new architecture, silent and efficient transportation, clean energy alternatives, The Venus Project's 25-acre design and research center and much more. Included are interviews with project founder and director, Jacque Fresco, his assistant Roxanne Meadows and others.

A STUDY GUIDE is included with this video. Structured in three sections. *Running time approximately 53 minutes.*

Price $29.95 plus S&H

THE VENUS PROJECT: THE REDESIGN OF A CIULTURE

This introductory video of The Venus Project presents a future that is attainable today. It shows a view of the wonderful choices and limitless possibilities in The Venus Project: a bold new direction for humanity.

Running time approximately 20 minutes.

Price $ 18.95 plus S&H

CITIES IN THE SEA

This video takes one on a journey of tomorrow's cities in the sea. It presents Jacque Fresco's fantastic and imaginative designs and footage of numerous massive ocean structures that will monitor and protect the ocean environment while providing recreation, education and an enriching lifestyle for the occupants. This can be accomplished with the utmost concern for the marine environment and energy efficient systems.

Running time approximately 15 minutes.

Price $19.95 & S&H

BOOKS

THE BEST THAT MONEY CAN'T BUY: BEYOND POLITICS, POVERTY, & WAR

Published in 2001, 170 pages, softcover. Hundreds of books address technological change, business process management, human productivity, and environmental issues. Almost all overlook the major element in all these systems — human beings and their social structures and culture. This book offers a possible alternative and methods to consciously fuse all these elements to create a sustainable future for all our planetary inhabitants, as well as fundamental changes in the way we regard ourselves, one another and our world. 70 color photos of Fresco's original designs representing this possible future are also presented.

Price $24.95 & S&H

THE VENUS PROJECT: THE REDESIGN OF A CULTURE

Published in 1995, 56 pages, softcover. This book gives the reader an introductory overview of *The Venus Project's* aims and proposals. The informative text is accompanied by 70 wonderful color and B&W photos and illustrations of many of *The Venus Project's* futuristic designs and concepts, as well as the existing 25-acre design and research facility. This book is sure to provide the reader with not only a graphic visualization of the possibilities for a better tomorrow, but also with a sense of hope for the future of humankind in our technological age.

Price $18.95 & S&H

LECTURES AND SEMINARS

Jacque Fresco is available for lectures and seminars.

ORDER THROUGH OUR WEBSITE OR CONTACT:

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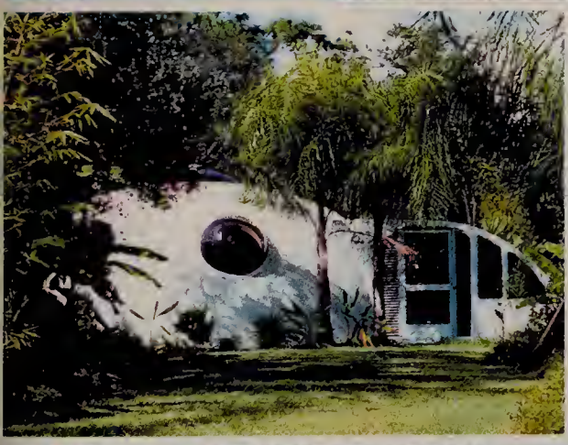
THE VENUS PROJECT'S RESEARCH CENTER













ABOUT THE AUTHOR



JACQUE FRESCO

JACQUE FRESCO

Futurist **Jacque Fresco** is a forerunner in the field of Industrial Design and Human Factors Engineering. Over the years, he has conceived a vast array of creative and innovative designs and plans for such things as prefabricated houses, automobiles, electronic and medical equipment, and hundreds of commercial products and inventions. Included among his many inventions are a radical aircraft wing structure and technique for viewing three-dimensional motion pictures without the need for special viewing glasses. In addition, he has served as technical advisor for a number of motion pictures. His works and ideas have been presented on numerous television and radio talk shows throughout the world, and articles about him have been featured in many national and international magazines and newspapers. Not only does he write and lecture about the future, but he actually lives in a future-oriented environment with his associate **Roxanne Meadows**. Together they have constructed a 25-acre research and development center in Venus, Florida where the future is unfolding. The Venus Project reflects the culmination of his life's work: the integration of the best of science and technology into a comprehensive plan for a new society based on human and environmental concern. It is a global vision of hope for the future of humankind in a technological age.



ROXANNE MEADOWS

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BY JACQUE FRESCO

THE BEST THAT MONEY CAN'T BUY

BEYOND POLITICS, POVERTY, & WAR



Fresco envisions a global civilization in which science and technology are applied with human and environmental concern to secure, protect, and encourage a more humane world for all people. This book offers a possible way out of our recurring cycles of boom and recession, famine, poverty, a declining environment, and territorial conflicts where peace is merely the interval between wars. It outlines an attainable, humane social design of the near future where human rights are no longer paper proclamations but a way of life. *The Best That Money Can't Buy* is a challenge to all people to work toward a society in which all of the world's resources become the common heritage of all of the earth's people. It is accompanied by 70 color photos of Fresco's original designs, which illuminate the fulfilling lifestyle of a global, resource-based economy.

GLOBAL CYBERVISIONS

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I must admit that I'm impressed — the book is b[illegible]ou the best of l[illegible]

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Original and holistic, this book stirs the mind, the heart, and the soul with its unique combination of vision and pragmatics; few volumes come close to it in breadth, depth, and the power to stir us to action…

— Arthur Shostak, Ph.D., Professor of Sociology, Drexel University

When too many were decrying technological evils and global doom, Jacque was a voice of reason pointing to a better world via humane applications of technology and a more appropriate social design. He is still that voice, so clearly and beautifully articulated in this book …

— Paul G. Hewitt, Author of *Conceptual Physics*

The Best That Money Can't Buy is a prophetic humanitarian blueprint for a new world for all the peoples of planet Earth for the 21st century. I consider this non-fictional watershed work the single most important book I have read in my lifetime. If you read only one book this century, invest your time and money in *The Best That Money Can't Buy* …

— Forry Ackerman, Mr. Science Fiction

America needs a new frontier: this work of love and genius by Jacque Fresco and Roxanne Meadows, outlines this new frontier — a frontier of the mind that opens us all to the astounding new possibilities that today's technology can provide. This book should be in every school in the country to inspire, motivate and alert all students and their families to what tomorrow can bring …

— Frank Ogden, known as Dr. Tomorrow, the world's foremost futurist.

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